

Expected Utility

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Random Variables and Expected Value

- We now move from probability to **decisions**.
- The key question: when outcomes depend on uncertain facts, how should we choose?
- Today we introduce the core tool: **expected utility**.

Associated Reading

Odds and Ends, Chapters 11 and 12

- A **random variable** is a variable that takes different numerical values in different states.
- In other words: a function from possibilities to numbers.

Examples:

- X = the age of the next US president at inauguration.
- Y = the number of children you will have in your lifetime.
- Z = the number of your friends who correctly predict tonight's game.

An Example

- 12 of your friends predicted the Lakers will win; 7 predicted the Clippers.
- Let X = the number of friends who predicted correctly.

$$X = \begin{cases} 12 & \text{if Lakers win} \\ 7 & \text{if Clippers win} \end{cases}$$

Given a random variable X and a probability function $\mathbf{Pr}$:

$$\text{Exp}(X) = \sum_i x_i \cdot \mathbf{Pr}(X = x_i)$$

This is a **weighted average** of the possible values, where the weights are the probabilities.

If $\Pr(\text{Lakers win}) = 0.7$:

$$\begin{aligned}\text{Exp}(X) &= 12 \times 0.7 + 7 \times 0.3 \\ &= 8.4 + 2.1 \\ &= 10.5\end{aligned}$$

Note: the “expected value” doesn’t have to be a value that X can actually take. It’s an average, not a prediction.

Decision Tables

States and Choices

We want to model decisions where outcomes depend on external facts outside our control.

	State 1	State 2
Choice 1	a	b
Choice 2	c	d

- **Choices:** the options available to you.
- **States:** ways the world might be that affect the outcome.
- **Outcomes:** a choice plus a state determines an outcome.
- **Utilities:** numbers representing how good each outcome is.

An Example

A friend has to choose between watching the Packers game and finishing a paper due Monday.

	Packers Win	Packers Lose
Watch Football	4	1
Work on Paper	3	2

The numbers reflect preferences. Higher is better.

Dominance

The simplest rule: **never choose a dominated option.**

- **A strongly dominates B :** A gives a strictly better outcome than B in every state.
- **A weakly dominates B :** A gives at least as good an outcome as B in every state, and strictly better in at least one.

A Case Where Dominance Applies

	Packers Win	Packers Lose
Watch Football	3	1
Work on Paper	4	2

Working on the paper gives a better outcome **regardless** of what the Packers do. It strongly dominates watching football.

This is compelling reasoning: if one option is better in every possible scenario, you should choose it.

Why Dominance Isn't Always Enough

	Packers Win	Packers Lose
Watch Football	4	1
Work on Paper	3	2

Neither option dominates the other. What to do depends on **how likely** each state is and **how much** you care about each outcome.

Consider two friends facing the “same” decision table (same ranking of outcomes):

- Chris’s cheap airline has a luggage problem in bad weather. Bags arrive a day late.
- Robin’s cheap airline has a structural problem in bad weather. It crashes.

Clearly these are different decisions, even though the *ranking* of outcomes is identical. The **magnitude** of the difference between outcomes matters too.

Maximising Expected Utility

*The rational choice is the one that **maximises expected utility**.*

Expected utility of a choice = weighted average of utilities of outcomes, weighted by probabilities:

$$\text{Exp}(U(\text{Choice})) = \sum_i \text{Pr}(\text{State}_i) \times U(\text{Choice}, \text{State}_i)$$

The Airline Example

	Good Weather (Pr = 0.8)	Bad Weather (Pr = 0.2)
Fly Cheap Airline	10	0
Fly Good Airline	6	5

$$\text{Exp}(\text{Cheap}) = 0.8 \times 10 + 0.2 \times 0 = 8.0$$

$$\text{Exp}(\text{Good}) = 0.8 \times 6 + 0.2 \times 5 = 5.8$$

Fly cheap. It has higher expected utility.

What If Bad Weather is Worse?

	Good Weather (Pr = 0.8)	Bad Weather (Pr = 0.2)
Fly Cheap Airline	10	-20
Fly Good Airline	6	5

$$\text{Exp}(\text{Cheap}) = 0.8 \times 10 + 0.2 \times (-20) = 4.0$$

$$\text{Exp}(\text{Good}) = 0.8 \times 6 + 0.2 \times 5 = 5.8$$

Now fly the reliable airline. The downside risk changed everything.

What If the Gap is Small?

	Good Weather (Pr = 0.8)	Bad Weather (Pr = 0.2)
Fly Cheap Airline	10	0
Fly Good Airline	9	8

$$\text{Exp}(\text{Cheap}) = 0.8 \times 10 + 0.2 \times 0 = 8.0$$

$$\text{Exp}(\text{Good}) = 0.8 \times 9 + 0.2 \times 8 = 8.8$$

The reliable airline wins. There's much less to gain by going cheap.

What If Bad Weather is Likely?

	Good Weather (Pr = 0.3)	Bad Weather (Pr = 0.7)
Fly Cheap Airline	10	0
Fly Good Airline	6	5

$$\text{Exp}(\text{Cheap}) = 0.3 \times 10 + 0.7 \times 0 = 3.0$$

$$\text{Exp}(\text{Good}) = 0.3 \times 6 + 0.7 \times 5 = 5.3$$

Bad weather being likely makes the reliable airline the clear choice.

The correct choice depends on all three of:

1. **Probabilities:** How likely are the different states?
2. **Upside:** How much do you gain by the cheaper/riskier option when things go well?
3. **Downside:** How much do you lose when things go badly?

Expected utility theory is the formal framework that integrates these three factors.

Worked Problems

A game is **fair** when neither player has an advantage. Both have expected payoff of zero.

If you pay some amount x to play a game with prizes, the game is fair when:

$$\text{Exp}(\text{your net payoff}) = 0$$

This is the same expected utility calculation, but now we're solving *backwards*: instead of computing expected utility given the numbers, we're finding the number that makes expected utility equal zero.

Problem 16: A Coin-Flip Game

A fair coin is flipped up to three times:

Outcome	Probability	Payoff
H (first flip)	$1/2$	+\$2
TH (second flip)	$1/4$	+\$40
TTH (third flip)	$1/8$	+\$800
TTT	$1/8$	-\$ x

Question: What must x be for the game to be fair?

Problem 16: Solution

Set expected payoff equal to zero:

$$\frac{1}{2}(2) + \frac{1}{4}(40) + \frac{1}{8}(800) + \frac{1}{8}(-x) = 0$$

$$1 + 10 + 100 - \frac{x}{8} = 0$$

$$111 = \frac{x}{8} \implies x = \$888$$

Notice the pattern in the expected contributions: 1, 10, 100. Each outcome is rarer but pays proportionally more. Each contributes exactly 10 times as much as the previous. The total expected winnings of \$111 must be offset by the losing outcome.

Problem 17: A Dice Game

A fair die is rolled up to twice:

Outcome	Probability	Payoff
1 on roll 1	$1/6$	$+\$x$
Not 1, then 2 on roll 2	$5/6 \times 1/6 = 5/36$	$+\$6x$
Not 1, then not 2	$5/6 \times 5/6 = 25/36$	$-\$6$

Question: What must x be for the game to be fair?

Problem 17: Solution

Set expected payoff equal to zero:

$$\frac{1}{6}x + \frac{5}{36}(6x) + \frac{25}{36}(-6) = 0$$

$$\frac{6x}{36} + \frac{30x}{36} - \frac{150}{36} = 0$$

$$\frac{36x}{36} = \frac{150}{36} \implies x = \frac{25}{6} \approx \$4.17$$

The algebra is clean even though the answer isn't a round number. Notice what's doing the work: $\frac{1}{6}x + \frac{5}{36}(6x)$ simplifies to just x , because the x and $6x$ prizes are weighted so that the two winning branches together contribute exactly x in expectation.

- We will look at the relationship between **money and utility**.
- Why isn't doubling your money the same as doubling your utility?
- And what does this tell us about insurance, gambling, and risk?